

RACHEL V. MEYER

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EDUCATION

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| 2021 - present | University of Florida (UF)
Ph.D. Candidate in Linguistics
Advisor: Ratree Wayland, Ph.D.
Dissertation: <i>Neural Network Analysis of Consonant Cluster Repair in Arabic Dialects</i>
Anticipated graduation date: August 2026 |
| 2017 - 2021 | The Ohio State University (OSU)
B.S. in Psychology and Linguistics, <i>summa cum laude</i>
Minor in Medieval and Renaissance Studies
Honors Research Distinction
Advisor: Rebeka Campos-Astorkiza, Ph.D. |

RESEARCH INTERESTS

Phonetics and Phonology

- Second language acquisition of phonetics and phonology
- Arabic dialects in phonological theory
- Sonority sequencing, consonant clusters, and syllable structures of Arabic
- Clinical speech in neurodegenerative diseases, speech as a biomarker/diagnostic tool

Computational Linguistics

- Feature theory and feature prediction in computational contexts
- Operationalizing acoustic correlates
- Expanding computational tools to dialectal Arabic and other low- and mid-resource languages

RESEARCH EXPERIENCE

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| 2021 - present | Speech, Lexicon, and Modeling Lab (UF)
Supervisors: Ratree Wayland, Ph.D and Kevin Tang, Ph.D |
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- Research Assistant for Smart Electropalatography for Linguistic and Medical Applications (SELMA) project (2023 - 2025)
 - Tested and analyzed electropalatography data
 - Obtained, cleaned, and analyzed corpus data
 - Trained, tested, evaluated, and improved neural network models
 - Performed statistical analysis, most commonly linear mixed-effects models
 - Co-wrote and edited six papers and four conference proceedings
 - Presented regularly at conferences

- 2020 - 2021 **Language Perception Lab (OSU)**
Supervisor: Mark Pitt, Ph.D
- Ran perception experiment participants while observing COVID safety protocols
 - Received training in data analysis, acoustic analysis, and experiment design
- 2019 - 2021 **Broad-coverage semantic models of human sentence processing (OSU)**
Supervisor: William Schuler, Ph.D
- Prepared training data
 - Marked semantic properties (coreference and scope) of nouns in Simple English Wikipedia articles
- 2019 - 2020 **Computational Experimental Phonology Lab (OSU)**
Supervisor: Becca Morley, Ph.D.
- Segmented audio files from experiment participants for a project on vowel length and consonant interaction
- 2018 - 2020 **Examining changes in social and communication skills in young children with Autism Spectrum Disorder participating in Early Intensive Behavior Intervention**
Supervisors: Kiwako Ito, Ph.D. and Elizabeth Kryszak, Ph.D
Nationwide Children's Hospital
- Transcribed audio of parents, therapists, and children interacting
 - Hand-corrected forced-aligned TextGrids from audio and transcripts
 - Marked turn-taking in transcripts

PUBLICATIONS

Journal Articles

- [1] R. Wayland, **R. Meyer**, and K. Tang, "Speech Markers of Parkinson's Disease: Phonological Features and Acoustic Measures," *Brain Sciences*, vol. 15, no. 11, p. 1162, 2025.
- [2] R. Wayland, **R. Meyer**, S. Vellozzi, and K. Tang, "Individual differences in language acquisition: The impact of study abroad on native English speakers learning Spanish," *Speech Communication*, vol. 174, p. 103 301, 2025, ISSN: 0167-6393.
- [3] R. Wayland, **R. Meyer**, R. Reddy, K. Tang, and K. W. Hegland, "Quantifying Lenition as a Diagnostic Marker for Parkinson's Disease and Atypical Parkinsonism," *BioMedInformatics*, vol. 4, no. 4, pp. 2287–2305, 2024.
- [4] R. Wayland, **R. Meyer**, S. Vellozzi, and K. Tang, "Lenition in L2 Spanish: The Impact of Study Abroad on Phonological Acquisition," *Brain Sciences*, vol. 14, no. 9, p. 946, 2024.
- [5] R. Wayland, H. W. Kim, **R. Meyer**, and S. Chen, "Asymmetric Discrimination of Pitch Contours among Non-native Tone Listeners," *Language and Linguistics*, vol. 42, no. 2, pp. 1–27, 2024.
- [6] A. Shamsi, M. Masapollo, **R. Meyer**, and R. Wayland, "Effects of motor practice on the temporal coordination of articulatory movements for non-native onset clusters: Kinematic and acoustic evidence using electromagnetic articulography," *The Journal of the Acoustical Society of America*, vol. 155, no. 3, A337–A338, 2024.

Conference Proceedings

- [1] N. Venkateswaran, **R. Meyer**, and R. Wayland, “Measuring the impact of segmental deviation on perceptions of accentedness using gradient phonological class features,” in *Proceedings of the Society for Computation in Linguistics (SCiL)*, vol. 8, Society for Computation in Linguistics, 2025.
- [2] **R. Meyer**, R. Wayland, K. Tang, S. Vellozzi, and R. Sengupta, “Measuring second language acquisition of Spanish lenition,” in *Proceedings of the Society for Computation in Linguistics (SCiL)*, vol. 7, Society for Computation in Linguistics, 2024, pp. 297–301.
- [3] R. Wayland, K. Tang, F. Wang, S. Vellozzi, **R. Meyer**, and R. Sengupta, “Neural network-based measure of consonant lenition in L2 Spanish,” in *Proceedings of Meetings on Acoustics*, vol. 52, AIP Publishing, 2023.
- [4] R. Wayland, K. Tang, F. Wang, S. Vellozzi, **R. Meyer**, and R. Sengupta, “Neural network-based measure of consonant lenition in Parkinson’s Disease,” in *Proceedings of Meetings on Acoustics*, vol. 52, AIP Publishing, 2023.

Conference Presentations

- [1] **R. Meyer**, “Toward customizable forced alignment for dialectal Arabic speech,” Talk given at the 39th Annual Symposium on Arabic Linguistics. Bloomington, Indiana, 2026.
- [2] N. Venkateswaran, **R. Meyer**, and R. Wayland, “Using self-supervised speech representations to track Parkinson’s Disease severity,” Poster presented at the 189th Meeting of the Acoustical Society of America. Honolulu, Hawaii, 2025.
- [3] **R. Meyer**, R. Wayland, and K. Tang, “Contrastive and privative featurization in phonological feature learning,” Talk given at Florida Linguistics Yearly Meeting. Gainesville, Florida, 2025.
- [4] R. Wayland, **R. Meyer**, K. Tang, S. Vellozzi, and R. Sengupta, “Neural network-based measure of voice quality in Parkinson’s disease,” Poster presented at the 188th Meeting of the Acoustical Society of America. New Orleans, Louisiana, 2025.
- [5] N. Venkateswaran, **R. Meyer**, and R. Wayland, “Quantifying the effects of the Indian English phonological system on accent perception using deep neural networks,” Poster presented at the 188th Meeting of the Acoustical Society of America. New Orleans, Louisiana, 2025.
- [6] A. Shamsi, **R. Meyer**, and R. Wayland, “Beyond the binary: Investigating gradient palatalization in Russian with deep learning models,” Poster presented at the 188th Meeting of the Acoustical Society of America. New Orleans, Louisiana, 2025.
- [7] H. Dalvi, **R. Meyer**, and R. Wayland, “Monophthongization of diphthongs in Southern American English: A perception study,” Poster presented at the 188th Meeting of the Acoustical Society of America. New Orleans, Louisiana, 2025.
- [8] N. Venkateswaran, **R. Meyer**, and R. Wayland, “Measuring the impact of segmental deviation on perceptions of accentedness using gradient phonological class features,” Poster presented at Society for Computation in Linguistics. Eugene, Oregon, 2025.
- [9] R. Wayland, **R. Meyer**, S. Vellozzi, and K. Tang, “Individual differences in L2 acquisition of Spanish voiced stop lenition,” Poster presented at Florida Psycholinguistics Meeting. Tallahassee, Florida, 2025.
- [10] N. Venkateswaran, R. Wayland, and **R. Meyer**, “Quantifying degrees of accentedness using first language (L1) phonological features,” Poster presented at New Ways of Analyzing Variation 51. Miami Beach, Florida, 2024.

- [11] **R. Meyer**, R. Wayland, K. Tang, S. Vellozzi, and R. Sengupta, “Measuring second language acquisition of Spanish lenition,” Poster presented at New Ways of Analyzing Variation 51. Miami Beach, Florida, 2024.
- [12] **R. Meyer**, R. Wayland, K. Tang, S. Vellozzi, and R. Sengupta, “Measuring second language acquisition of Spanish lenition,” Talk given at Society for Computation in Linguistics. Irvine, California, 2024.
- [13] A. Shamsi, M. Masopollo, **R. Meyer**, and R. Wayland, “Speech motor sequence learning reflects changes in processing load, not unfamiliar articulatory coordination,” Poster presentation at the 186th Meeting of the Acoustical Society of America. Ottawa, Canada, 2024.
- [14] R. Wayland, K. Tang, **R. Meyer**, S. Vellozzi, and R. Sengupta, “Neural network-based measure of voice quality in Parkinson’s Disease,” Talk given at Florida Linguistics Yearly Meeting. Gainesville, Florida, 2024.
- [15] R. Wayland, K. Tang, **R. Meyer**, S. Vellozzi, and R. Sengupta, “Impact of electropalatography on speech quality: A deep neural approach,” Talk given at Florida Linguistics Yearly Meeting. Gainesville, Florida, 2024.
- [16] R. Wayland, K. Tang, F. Wang, S. Vellozzi, **R. Meyer**, and R. Sengupta, “Neural network-based measure of consonant lenition in L2 speech,” Poster presentation at the 185th Meeting of the Acoustical Society of America. Sydney, Australia, 2023.
- [17] R. Wayland, K. Tang, F. Wang, S. Vellozzi, **R. Meyer**, and R. Sengupta, “Neural network-based measure of consonant lenition in Parkinson’s Disease,” Poster presentation at the 185th Meeting of the Acoustical Society of America. Sydney, Australia, 2023.
- [18] **R. Meyer**, S. Vellozzi, R. Wayland, K. Tang, and R. Sengupta, “Posterior probability as a measure of lenition in L2 Spanish,” Poster presentation at Florida Psycholinguistics Meeting. Gainesville, Florida, 2023.
- [19] A. Shamsi, M. Masopollo, **R. Meyer**, and R. Wayland, “Motor practice stabilizes the temporal coordination of articulatory movements for non-native onset clusters,” Poster presented at the 2023 Boston Speech Motor Control Symposium. Boston, MA, 2023.
- [20] **R. Meyer**, “Consonant sequence palatalization in media Arabic,” Talk given at the 36th Annual Symposium in Arabic Linguistics. Amherst, Massachusetts, 2023.
- [21] **R. Meyer**, R. Wayland, and F. Wang, “Measuring lenition in second language Spanish,” Poster presentation at the 183rd Meeting of the Acoustical Society of America. Nashville, Tennessee, 2022.
- [22] **R. Meyer**, “Coarticulation in second language Spanish: Learning to connect sounds in a second language,” Talk given at Current Approaches in Spanish and Portuguese Second Language Phonology. San Jose, California, 2020.
- [23] **R. Meyer**, “Coarticulation in second language Spanish: Learning to connect sounds in a second language,” Poster presentation at The Ohio State University Undergraduate Research Symposium. Columbus, Ohio, 2019.

TEACHING EXPERIENCE

2025 - present

Teaching Assistant

Introduction to Linguistics

- Graded homework, quizzes, and exams for approximately 65 students
- Held office hours for in-person and online sections

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| 2022 - 2023 | Teaching Assistant
Sounds of Human Language (Introductory Phonetics) |
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- Organized and led recitation small groups for approximately 40 students per semester
 - Designed recitation materials
 - Graded homework and quizzes
 - Held office hours

SERVICE

Conference Organization

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| 2024 - present | Co-organizer, Phonetics and Phonology Reading Group |
| 2024 - 2025 | Organizing Committee, Florida Linguistics Yearly Meeting |
| 2023 | Volunteer, Florida Psycholinguistics Meeting |

Reviewing

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| 2025 - present | Asian-Pacific Journal of Second and Foreign Language Education |
| 2024 - 2025 | Florida Linguistics Yearly Meeting |

Outreach

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| 2025 | Cade Museum for Creativity and Invention Language Day |
| 2022 - 2025 | Treasurer, Graduate Linguistics Society |

SCHOLARSHIPS

Assistantships

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| 2025 - present | Graduate Teaching Assistantship |
| 2023 - 2025 | Graduate Research Assistantship |
| 2022 - 2023 | Graduate Teaching Assistantship |

Research and Conference Funding

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| 2025 | College of Liberal Arts and Sciences travel award, \$500 |
| 2025 | Graduate Student Council travel award, \$400 |
| 2024 | College of Liberal Arts and Sciences travel award, \$300 |
| 2023 | College of Liberal Arts and Sciences travel award, \$300 |
| 2020 | Eleanor Ruffing McMahon Award for Conference Travel, \$750 |
| 2020 | Undergraduate Research Scholarship, \$2500 |

Merit Scholarships

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| 2021 - 2022 | Linguistics Funding Award |
| 2017 - 2021 | Maximus Scholarship, semesterly, \$2500 |

SKILLS

Languages

English (native)

Arabic (Modern Standard-intermediate, Levantine-intermediate low)

French (intermediate)

Spanish (working)

Computational Tools

Python, R, Praat, SQL

Experience with Tensorflow, Keras